# Triadic Visualization of Deutsch's Algorithm: Circuit, ZX–Calculus, and Phasor–Fill Approaches

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***Abstract*—Deutsch's algorithm solves the oracle problem of determining whether a Boolean function $f : \{0, 1\} \to \{0, 1\}$ is constant or balanced with a single quantum query—contrasting with the two queries required classically. I present three complementary perspectives: the standard circuit-level implementation that makes explicit the reduction in query complexity; a ZX–calculus diagrammatic derivation where spider-fusion and phase-elimination collapse the oracle–Hadamard pattern in one rewrite; and a novel 2-D phasor–fill representation. In the latter, each qubit is drawn as a unit circle whose external arrow encodes equatorial phase and whose top/bottom fill encodes positive/negative imaginary phase, so that Deutsch's algorithm appears simply as successive arrow rotations on an empty (real-only) circle. Together, these views cater to implementers, categorical/diagrammatic reasoners, and educators seeking intuitive, whiteboard-friendly depictions, enriching both the analysis and communication of quantum speedups.**

***Index Terms*—Quantum computing, Deutsch's algorithm, ZX-Calculus, quantum circuit analysis, quantum interference.**

## I. Introduction

Deutsch's problem asks: given a black-box Boolean function $f : \{0, 1\} \to \{0, 1\}$, decide whether $f$ is *constant* ($f(0) = f(1)$) or *balanced* ($f(0) \neq f(1)$). Classically, a deterministic algorithm requires two evaluations of $f$; Deutsch's quantum algorithm demonstrates a provable advantage by solving the problem with a single query using superposition and interference.

In this paper I present three complementary perspectives on Deutsch's algorithm:

1) **Classical circuit approach (Section II).** I recall the standard two-query classical circuit and then show the quantum circuit that reduces the query count to one, with a detailed low-level step-by-step explanation.
2) **ZX–calculus approach (Section III).** I translate the Deutsch circuit into a ZX diagram of $Z$ and $X$ spiders, and demonstrate how a single application of the spider-fusion and phase-elimination rewrite rules collapses the Hadamard–oracle–Hadamard pattern in one step.
3) **Phasor–fill representation (Section IV).** I introduce a novel 2-D depiction of a qubit as a unit circle where:
   - An *external arrow* around the rim encodes the equatorial phase $\phi$ (up = $|0\rangle$, right = $|-\rangle$, down = $|1\rangle$, left = $|+\rangle$).
   - A *top/bottom fill* of the circle encodes any imaginary-phase component ($+i$ vs. $-i$), while an empty circle indicates a purely real state.

   In this picture, each single-qubit gate is simply a rotation of the arrow, and Deutsch's algorithm appears as successive arrow rotations on an empty (real-only) circle.

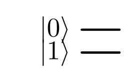
The remainder of the paper is organized as follows. Section II covers the classical and quantum circuit implementations; Section III gives the ZX–calculus derivation; Section IV develops the phasor–fill representation and visualizes Deutsch's algorithm within it; and Section V concludes with comparisons and future directions.

## II. Classical Approach

Deutsch's algorithm in the standard circuit picture proceeds in five concrete steps. Below I give both the mathematical/transformation description and the corresponding circuit diagram for each step.

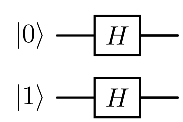
1) **Initialization.**

$$|\psi_0\rangle = |0\rangle \otimes |1\rangle .$$



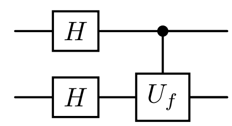

2) **First Hadamard layer.** Apply $H$ to each qubit:

$$|\psi_1\rangle = (H \otimes H) |\psi_0\rangle = \tfrac{1}{2}(|0\rangle + |1\rangle) \otimes (|0\rangle - |1\rangle).$$




3) **Oracle call.** Execute $U_f \colon |x\rangle |y\rangle \mapsto |x\rangle |y \oplus f(x)\rangle$, imprinting a phase $(-1)^{f(x)}$ on the first qubit:

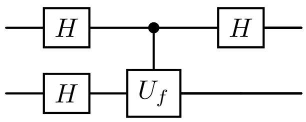
$$|\psi_2\rangle = U_f \ |\psi_1\rangle = \frac{1}{2} \sum_{x=0}^{1} (-1)^{f(x)} |x\rangle \otimes (|0\rangle - |1\rangle).$$




4) **Second Hadamard on first qubit.** Apply $H$ to the first wire to interfere the phases:

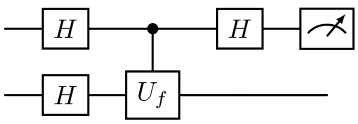
$$|\psi_3\rangle = (H \otimes I) |\psi_2\rangle = \frac{1}{2\sqrt{2}} \sum_{x,y=0}^{1} (-1)^{f(x)+xy} |y\rangle \otimes (|0\rangle - |1\rangle).$$

The amplitude on $|1\rangle$ of the first qubit is proportional to $f(0) \oplus f(1)$.




5) **Measurement.** Measure the first qubit in the computational basis:

$$\begin{cases} |0\rangle & \Rightarrow \textit{constant}, \\ |1\rangle & \Rightarrow \textit{balanced}. \end{cases}$$

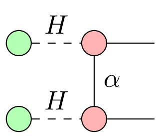



## III. ZX–Calculus Approach

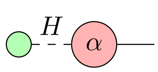
In ZX–calculus I redraw the same five steps as a network of $Z$ (green) and $X$ (red) spiders, with dashed edges for Hadamards. Each rewrite step corresponds exactly to one layer of the standard circuit.

1) **Initial ZX diagram.** Translate the post-Hadamard state and oracle into spiders and a phase gadget $\alpha = f(0) \oplus f(1)$:

$$\text{Green spiders at inputs} \xrightarrow{H} \text{Red spiders,}$$
$$\text{Oracle } U_f \cong \text{controlled-phase } \alpha.$$

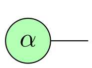



2) **Spider fusion.** Fuse same-colored spiders connected by plain wires, summing their phases. The two green spiders merge into one; the two red into one carrying phase $\alpha$.

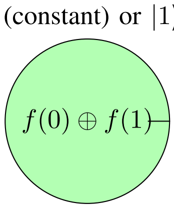



3) **Phase elimination.** Apply the rule that a red spider with zero phase disappears (identity), and with phase $\pi$ becomes a $\pi$-phase on the green spider. Thus:

$$\alpha = \begin{cases} 0 & \Rightarrow \text{constant,} \\ \pi & \Rightarrow \text{balanced.} \end{cases}$$



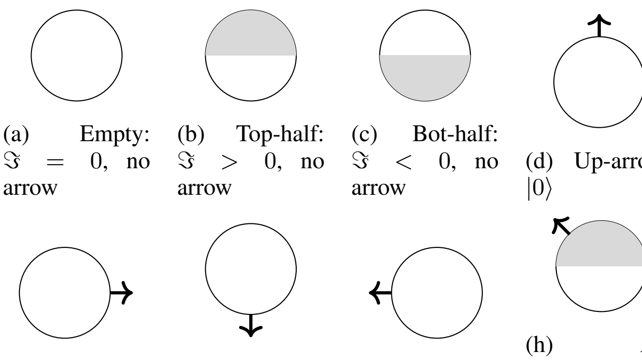

4) **Final read-out.** The remaining green spider with no phase measures $|0\rangle$ (constant) or $|1\rangle$ (balanced):




## IV. Phasor–Fill Representation Approach

Before using it to visualize Deutsch's algorithm, I first explain my 2-D "phasor-fill" picture of a qubit, where the circle's fill encodes the imaginary component and an external arrow encodes the equatorial phase.

### A. Representation Primitives



(a) Empty: $\Im = 0$, no arrow (b) Top-half: $\Im > 0$, no arrow (c) Bot-half: $\Im < 0$, no arrow (d) Up-arrow: $|0\rangle$

(e) Right-arrow: $|-\rangle$ (f) Down-arrow: $|1\rangle$ (g) Left-arrow: $|+\rangle$ (h) Arrow+shading: mixed state

Fig. 1: Phasor–fill primitives: (a) empty circle, (b) top-half fill, (c) bottom-half fill, (d) up-arrow, (e) right-arrow, (f) down-arrow, (g) left-arrow, (h) shading + arrow.

### B. Visualizing Deutsch's Algorithm

Since Deutsch's gates only introduce real $\pm 1$ phases, the circles remain empty; only the arrow orientation changes.

1) **Initialization.**

$$|\psi_0\rangle = |0\rangle \quad \longrightarrow \quad \text{Up-arrow, empty circle}$$

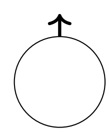


Fig. 2: Initialization: $|0\rangle$

2) **First Hadamard.**

$$|\psi_1\rangle = H\,|0\rangle = |+\rangle \quad \longrightarrow \quad \text{Left-arrow, empty circle}$$

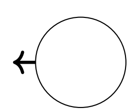


Fig. 3: After first Hadamard: $|+\rangle$

3) **Oracle Phase Kick.**

$$|\psi_2\rangle = Z^{f(0)\oplus f(1)}\,|+\rangle$$

$$\longrightarrow \begin{cases} \text{Left-arrow, empty} & f \text{ constant,} \\ \text{Right-arrow, empty} & f \text{ balanced.} \end{cases}$$

(a) Constant $f$ (b) Balanced $f$

Fig. 4: After oracle: arrow flips only if $f$ is balanced.

4) **Second Hadamard.**

$$|\psi_3\rangle = H\,Z^{f(0)\oplus f(1)}\,|+\rangle$$

$$\longrightarrow \quad \begin{cases} \text{Up-arrow, empty} & f \text{ constant,} \\ \text{Down-arrow, empty} & f \text{ balanced.} \end{cases}$$

(a) Constant (b) Balanced

Fig. 5: After second Hadamard: up vs. down arrow indicates constant vs. balanced.

5) **Measurement.** Reading "up" as $|0\rangle$ (constant) or "down" as $|1\rangle$ (balanced) completes the one-query decision.

## V. Conclusion

In this paper I have presented three complementary ways to understand Deutsch's algorithm. First, the *classical circuit* approach shows directly why two oracle queries are required deterministically, and how the quantum circuit reduces this to one by exploiting superposition and interference (Section II). Second, the *ZX–calculus* approach recasts the same one-query speedup as a concise diagram rewrite, where spider-fusion and phase-elimination collapse the oracle–Hadamard–Hadamard pattern in one step (Section III). Third, my new *phasor–fill representation* flattens the qubit onto a 2-D circle: an external arrow encodes equatorial phase, and the circle's fill level encodes any imaginary component (Section IV).

Applied to Deutsch's algorithm, the phasor–fill picture tracks the first qubit through initialization, Hadamard, oracle kick, second Hadamard, and measurement merely by rotating its arrow around an empty (real-only) circle. No higher-dimensional intuition is required—each gate becomes a simple rotation, and the final up/down orientation immediately signals constant vs. balanced.

Together, these views reinforce the same underlying quantum speedup while catering to different audiences and use cases:

- The circuit description is useful for implementation and for comparing quantum vs. classical query complexity.
- The ZX–calculus derivation appeals to categorical and diagrammatic reasoning, making equational proofs of optimization transparent.
- The phasor–fill representation offers a pedagogical shorthand ideal for teaching and for rapid white-board sketches in two dimensions.

Future work includes extending the phasor–fill method to algorithms that introduce genuine $\pm i$ phases (e.g. phase-estimation), and integrating it into interactive visualization tools. I believe that, taken together, these three approaches provide a robust toolkit for both analyzing and communicating quantum algorithms at multiple levels of abstraction.

## References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